

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-003 - Lakwa-003

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000028
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Lakadong+Therria (LKT)
Spud / Completion	2020-10-09 / 2020-12-07
Measured / TVD	3482 m / 3455 m
Trajectory	Vertical; inc 3.6 deg; az 209.8 deg
Formation Interval	1853-3155 m
Lithology	Sandstone-coal-shale sequence
Pore Pressure Reference	3593 psi
Fracture Gradient	14.49 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Low (37.9/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	870	Surface
9 5/8 in casing	2019	Intermediate
7 in liner/casing	3064	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2020-11-05 1732 m	Cementing Issue	High	Displacement pressure became unstable. Mitigation: Control displacement rate and reconcile pumped/returned volumes. Lesson: Offset loss history should inform cementing hydraulics.
2020-11-17 2414 m	Cementing Issue	High	Cement returns were below planned percentage. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Plan conservative displacement rate in weak formations.
2020-10-21 882 m	Kick	Medium	Unexpected background gas rise preceded influx. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Treat pit gain plus gas increase as an early influx indicator.
2020-10-28 1272 m	Pressure Spike	Medium	Pressure trend exceeded recent moving baseline. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2020-11-13 2180 m	Fishing Operation	Medium	Normal pipe-recovery attempts were unsuccessful. Mitigation: Use approved fishing program with controlled overpull and circulation. Lesson: Capture retrieval outcome as institutional knowledge.
2020-11-15 2284 m	Torque Spike	Medium	High torque developed while WOB remained elevated. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Use offset torque trend to set adaptive operating envelope.
2020-11-16 2336 m	Mud Loss	Medium	Intermittent circulation losses developed while drilling ahead. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Track flow balance and pit volume trend before losses become severe.

Completion Summary

The synthetic well reached 3482 m MD in the Lakadong+Therria interval. The NWIS knowledge record links 11 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.